## Single Slit Diffraction

### 1. Definition

Single slit diffraction is the phenomenon in which light bends and spreads into the region of geometrical shadow when it passes through a narrow slit whose width is comparable to the wavelength of light.

### 2. Principle

Single slit diffraction is based on the wave nature of light. According to Huygens' principle, every point on a wavefront acts as a source of secondary wavelets. These wavelets interfere with one another and produce a diffraction pattern.

### 3. Experimental Arrangement

A monochromatic light source illuminates a narrow slit of width a. The diffracted light is observed on a screen placed at a distance D from the slit.

The pattern consists of a broad central bright maximum with alternate dark and weaker bright fringes on both sides.

### 4. Condition for Minima (Dark Fringes)

For destructive interference, the condition for minima is:

$a \sin \theta = n\lambda$

Where:

- a = width of the slit
- $\theta$ = angle of diffraction
- $\lambda$ = wavelength of light
- n = 1, 2, 3, ...

For small angles:

$\sin \theta \approx \tan \theta \approx y/D$

Therefore, the position of the nth minimum is:

$y_n = n\lambda D/a$

Here, $y_n$ is the distance of the nth minimum from the centre of the pattern.

### 5. Width of the Central Maximum

The first minima occur on either side of the central maximum at:

$y_1 = \lambda D/a$

Hence, the width of the central maximum is:

$W = 2\lambda D/a$

### 6. Characteristics of Single Slit Diffraction

1. The central maximum is the brightest and widest fringe.
2. The central maximum is twice as wide as each secondary maximum, approximately.
3. The intensity of secondary maxima decreases away from the centre.
4. The diffraction pattern is symmetrical about the central maximum.
5. The width of the central maximum increases with wavelength and screen distance.
6. The width decreases when the slit width increases.

**7. Difference Between Interference and Diffraction**

| **Interference** | **Diffraction** |
|---|---|
| Usually produced by waves from two coherent sources. | Produced by waves from different parts of the same slit. |
| Fringes are approximately equally spaced under standard conditions. | The central maximum is twice as wide as the distance between successive minima. |
| Bright fringes may have nearly equal intensity for equal-amplitude sources. | Secondary maxima have decreasing intensity. |

**8. Applications**

1. Used to study the wave nature of light.
2. Used to determine the wavelength of light.
3. Used to study the resolving power of optical instruments.
4. Used in spectroscopy and optical measurements.

**9. Conclusion**

Single slit diffraction demonstrates the wave nature of light. The central maximum is the brightest and widest part of the diffraction pattern.

Important formula:

$W = 2\lambda D/a$